Effect of homocysteine lowering treatment on cognitive function: a systematic review and meta-analysis of randomized controlled trials.

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Elevated total plasma homocysteine has been linked to the development of cognitive impairment and dementia in later life and this can be reliably lowered by the daily supplementation of vitamin B6, B12, and folic acid. We performed a systematic review and meta-analysis of 19 English language randomized, placebo-controlled trials of homocysteine lowering B-vitamin supplementation of individuals with and without cognitive impairment at the time of study entry. We standardized scores to facilitate comparison between studies and to enable us to complete a meta-analysis of randomized trials. In addition, we stratified our analyses according to the folate status of the country of origin. B-vitamin supplementation did not show an improvement in cognitive function for individuals with (SMD = 0.10, 95%CI -0.08 to 0.28) or without (SMD = -0.03, 95%CI -0.1 to 0.04) significant cognitive impairment. This was irrespective of study duration (SMD = 0.05, 95%CI -0.10 to 0.20 and SMD = 0, 95%CI -0.08 to 0.08), study size (SMD = 0.05, 95%CI -0.09 to 0.19 and SMD = -0.02, 95%CI -0.10 to 0.05), and whether participants came from countries with low folate status (SMD = 0.14, 95%CI -0.12 to 0.40 and SMD = -0.10, 95%CI -0.23 to 0.04). Supplementation of vitamins B12, B6, and folic acid alone or in combination does not appear to improve cognitive function in individuals with or without existing cognitive impairment. It remains to be established if prolonged treatment with B-vitamins can reduce the risk of dementia in later life.